The effect of gain adaptation on perception and posture*

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ABSTRACT

Consistent motion simulation in Virtual Reality (VR) applications is challenging due to constraints on tracking technologies and locomotion or interaction paradigms that scale motion. In this study, we measured (a) the point of subjective stationarity (PSS) during active self-motion, (b) postural sway in the dark, and (c) visually-induced postural sway during quiet stance, both before and after adapting to different gains between physical self-motion and the motion portrayed in the virtual environment. Participants adapted to each of three adaptation gain levels in separate blocks: normal, reduced, and increased, in which observers' physical motion was scaled and displayed as the virtual motion. We measured PSS during active self-motion and postural sway during quiet stance in both left-right and front-back directions in separate sessions. We found that the PSS measured during self-motion did not vary with adaptation gain. However, postural sway elicited by visual perturbation was modulated after adapting to non-unity gains. We also measured baseline postural sway prior to adaptation and found that exposure to virtual motion under unity (normal) gain increased the postural variability along the left-right direction, when tested without visual feedback (in dark). Collectively these results suggest that while observers adapt to gain, active self-motion provides sufficient somatosensory feedback to counteract the shift in perceived motion. As a result, PSS remains consistent across all gain manipulations in our experiment setup. In contrast, postural responses during quiet stance did not recalibrate immediately after motion gain perturbation was removed suggesting that they operate independently of perceptual mechanisms.

Introduction

In natural environments, sensory cues from the vestibular system (linear and angular acceleration) and the kinesthetic or proprioceptive system (limb and body position/orientation) are typically consistent with visual signals, providing complementary or redundant information[1]. However, self-motion perception in virtual reality (VR) presents unique challenges due to conflicts between visual and non-visual cues. Commonly, VR presents visual motion that is unaccompanied by, scaled, or delayed relative to physical motion due to design purposes (redirected walking, flight simulation), or hardware constraints (limited field of view, latency). This discrepancy, referred to as visual gain in the case of scaling, represents a mismatch between visually presented and physically executed movement, and can lead to perceptual errors, instability, and/or cybersickness[2–4]. Similar to depth and egocentric distance underestimations in VR, the perceptual biases for motion perception are also direction-dependent[5–8]. For motion-in-depth, users tend to exhibit perceptual and behavioral challenges in accurately interpreting dynamic depth cues compared to lateral motion in VR environments. Huang et al.[9] found that both the movement time and error rate in acquiring targets with motion-in-depth followed strong regularities, with performance significantly affected by visual factors such as shadows and speed. In contrast, lateral motion perception appears to be relatively preserved. For example, Jörges and Harris[10] demonstrated that participants' ability to discriminate the speed of lateral movement was not significantly biased between static and visually simulated lateral self-motion trials, though precision decreased slightly when self- and object motions were in opposite directions.

In the case of self-motion perception, Crowell et al.[11] showed that accurate perception of self-motion during active head turns relies on a combination of vestibular stimulation, neck proprioception, and efference copy. Self-motion perception during active head turns is therefore more accurate than during passive head turns due to this rich sensory integration. In line with Crowell et al.'s findings, Wexler and van Boxtel[12] showed that active engagement with the environment can enhance depth perception by providing additional sensory cues. They found that gain applied to active motion parallax systematically altered perceived depth and egocentric distance. In a previous study, we reported similar results and further found that modest

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gain changes on the order of 10% affected the perceived stationarity of the world[13]. Kim and Palmisano found that passive visual-vestibular conflicts generated stronger illusions of self-motion compared to active viewing conditions[14]. Together, these findings raise the question of whether adaptation to altered motion gain recalibrates the perception of self-motion.

Humans naturally recalibrate their movements, e.g., walking, turning, and reaching, to adapt to environmental changes such as unexpected acceleration or deceleration[15]. This recalibration is evident in head rotation and prism adaptation tasks under persistent sensorimotor distortions[16–18]. The relationship between self-motion perception and sensory integration has been extensively studied using both naturalistic observations[19] and virtual reality (VR) paradigms[20]. Studies of this have shown that perceptual and motor responses may adapt differently to visuomotor conflicts, especially when active movement and postural regulation are involved[21,22]. The point of subjective stationarity (PSS) is a key measure studied to understand the perceptual mechanisms involved in reconciling visuomotor conflicts. It defines the perceptual threshold at which the visual and kinesthetic motions are perceived as matched in a perceptually stable world. The perception of a stable environment supports an accurate and temporally consistent spatial representation of the world, allowing for reliable and repeatable perception and action relative to this persistent frame of reference. Dokka et al.[23] reported that observers showed flexible compensatory mechanisms in PSS, attributing visual-vestibular distortions between self-motion in a stable world or an unstable world at the cost of less accurate self-motion perception. This flexibility in multisensory integration allows adaptive calibration during motion changes to avoid disorientation or prevent falls. The classic 'moving room' paradigm by Lee and Lishman[24,25] revealed that moving visual environments in the absence of actual movement can elicit compensatory postural adjustments that led to the loss of balance or fall. The moving room study underscores the dominance of visual input in postural control, where the nervous system adjusts motor outputs to stabilize posture when visual and vestibular signals are misaligned[26]. Subsequent studies further confirmed that visually-induced self-motion can modulate standing balance control and postural sway[27,28]. In light of these findings, although a shift in PSS could signal rebalancing of perception to maintain appropriate self-motion perception and avoid disorientation, prevention of falls relies on visuomotor postural responses and modulation of postural reflexes[29,30].

Adaptation to gain requires a rebalancing of visual and non-visual self-motion signals and could be reflected in perceived stability, postural responses, or both. In this study, we examined the effect of adaptation to translational gain on: (i) the change in PSS under active self-motion in gain-manipulated virtual environments, which reflects how mismatched visual and motor signals recalibrate perception; and (ii) balance during quiet stance either in response to no visual input or visually-elicited perturbations. We predict that after prolonged exposure to unusual gain conditions, the PSS and the postural gain response to a visual perturbation will both scale in the direction of the adaptation gain, reflecting a new equilibrium for both perception and balance. We also predict that if the adaptation mechanism is common, then the PSS shift and the change in gain response to a perturbation will be highly correlated.

Experiment 1

Under some conditions, motion parallax can provide robust depth information comparable to that of binocular stereopsis[31]. The perceived stability of the world affects the ability to attribute motion parallax to depth rather than non-rigid scene and object motion[13,23]. When the world appears stationary and stable despite the presence of optic flow, then the motion in the visual field is perceived as self-motion and static depth in the scene. This experiment investigates how systematic gain manipulation on motion parallax affects the PSS, and whether observers adapt to such a manipulation.

This experiment contains two parts: we started with binocular viewing conditions (Experiment 1a), then further investigated the adaptation effect under monocular viewing (Experiment 1b).

Methods

Participants

Eighteen observers (thirteen female and five male, age range 22–38 years, mean = 28.2) were recruited from York University and received monetary compensation for their participation. All participants had normal or corrected to normal vision. Among them, ten were experienced psychophysical observers (including an author) and eight were novices. Prior to participation, they provided written informed consent. All experiments were approved by the York University Human Participants Review Committee (HPRC). The methods were performed in accordance with relevant guidelines and regulations and adhered to the principles outlined in the Declaration of Helsinki.

Apparatus

An Oculus Rift S was used to present PC-based WebXR and PsychoPy 2022 visual simulations over a QuestLink cable. The LCD displays have a resolution of 1280×1440 pixels per eye and a diagonal field of view of 115°. The headset has built-in cameras and an inertial measurement unit for 6 DoF (Degrees of Freedom) location/orientation tracking, which we used to manipulate the motion gain in real time. The frequency of motion tracking and synchronous frame rate was 60 Hz. The right-hand controller was used by the observers to calibrate the virtual environment (VE) and register their responses.

Design

This study used a within-subject factorial design. The independent variables were adaptation gain (0.667, 1.0, and 2.0), test gain (0.667, 0.8, 1.0, 1.25, 1.5, and 2.0) and viewing (binocular, monocular). The adaptation gains and viewing conditions were presented in separate blocks and all the test gains were assessed after each adaptation period. For the test gain, we used the method of constant stimuli and each level was repeated five times. The dependent variable included the discrimination response (more, less) and tracked head motion (location and orientation with time stamp).

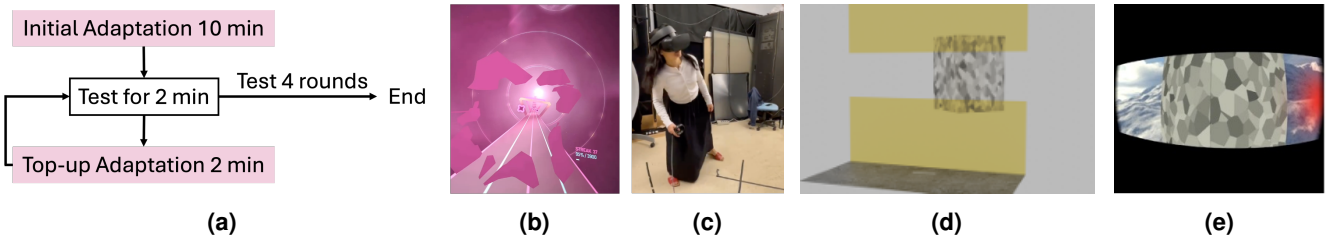


Figure 1. Procedure and stimuli used in Experiment 1. (a) diagram of the experiment procedure, (b), (c) in the adaptation phase, the observer is moving to the right to intersect the approaching target, (d) the 'fold' stimulus in the test phase, a vertical dihedral angle which was viewed through top and bottom apertures that were opaque black but are depicted as translucent here for illustrative purposes, (e) right eye's view in the test phase, red boundary light denotes that the movement has met the distance requirement for observer produced motion parallax to the right side.

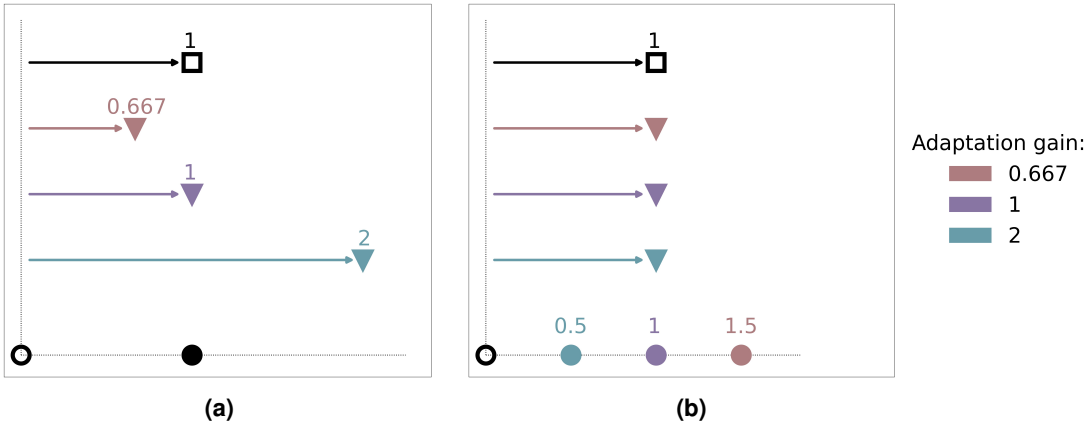


Figure 2. Top-down view of the spatial relations between the observer, virtual camera, and game target, with varying levels of motion gain. Hollow markers illustrate original positions of the observer (circles) and a visual target offset to the right by 1 unit distance (squares). Solid markers represent positions after the observer moves. (a) shows the scenario where the observer moves their body the same amount (black solid circle) as the apparent offset of the target; in this case, the virtual cameras (triangles) translate by their corresponding gains and render the observer's views according to the movement made. The motion will undershoot and overshoot the desired virtual alignment for the low and high gains, respectively. (b) shows the scenario where the observer aligns the virtual cameras with the target by performing the motions that correspond to the gain levels. Under gain of 1, to align with the visual target the observer should translate by the same distance of 1 unit (purple circle); while the observer is required to translate 1.5 unit of distance under gain of 0.667 and 0.5 unit of distance under gain of 2, respectively.

Procedure

The experiment consisted of an adaptation phase and a test phase. As shown in Figure 1a, observers started with an initial adaptation phase, where they were immersed in a VR game for 10 minutes. The adaptation scene was modified from the open-source MoonRider game (moonrider.xyz). During the adaptation phase, observers moved their bodies laterally to intersect flying targets, which "exploded" after colliding with the participant (Figures 1b, 1c, and Supplementary Video S1). Their virtual motion was scaled by three adaptation gains of reduced (0.667), normal (1.0), and increased (2.0) in separate blocks, depicted in Figure 2. Observers were required to achieve at least 90% task success rate on the number of targets intersected, so that the overall task demand was consistent across individuals. Upon finishing the initial adaptation, observers were transitioned into the test scene. This scene contained a 'fold' stimulus – a vertical dihedral angle with its top and bottom edges occluded by

apertures (Figure 1d). During the test phase, observers swayed left and right over 20 cm to a 0.5 Hz metronome beat, and their virtual motion was scaled by one of six test gains randomly selected from trial to trial: 0.667 ($\frac{2}{3}$), 0.8 ($\frac{4}{5}$), 1, 1.25 ($\frac{5}{4}$), 1.5 ($\frac{3}{2}$), and 2.0. Figure 1e shows the virtual view displayed to the right eye as the observer swayed to the right. Observers performed a 2-alternative forced choice (2AFC) task, where they discriminated whether the virtual motion appeared to be 'more or less than they physically moved' (Supplementary Video S2 and S3). Each segment of the test phase was 2 minutes; following this, observers were transitioned back to the adaptation phase for 2-minute top-ups followed by subsequent testing segments until they completed all four test segments. Each block took about 30 minutes, totaling 1.5 hours over the three blocks for each viewing condition. Head position and motion data from the 6-DOF headset head tracker were recorded continuously throughout this and subsequent experiments.

Results

Observers reliably discriminated the levels of gain tested (Figure 3). Figure 3a shows the proportion of 'more' responses as a function of the test gain obtained from a representative observer in Experiment 1a (binocular). The point of subjective equality (PSE, 50% point) and just noticeable difference (JND, 75% threshold) were estimated by fitting a sigmoidal cumulative normal probability distribution to each psychometric function of the corresponding adaptation block. The PSE represents the level of gain that the observer perceived, on average, as moving neither more nor less than their physical motion. Thus, it is the PSS after adaptation where the perceived virtual motion matches their physical motion. The JND in this experiment represents the observer's sensitivity to gain manipulations. Anecdotally, several observers reported mild motion aftereffects following each 30-minute block.

To test the hypotheses outlined above, one-way analyses of variance (ANOVA) were carried out on the dependent variables (DV) of the observers' PSS, JND, and head motion extent (the averaged distance traveled in a full cycle of left-right translation) with an alpha of 0.05:

$$D.V. \sim Adaptation + Error(Observer/Adaptation) \tag{1}$$

For Experiment 1a, these analyses showed that there was a marginally significant effect of adaptation gain on PSS, but the effect size was very small (Table 1). No significant main effect of adaptation gain was found for JND or head motion extent. Pairwise t-tests with Bonferroni correction suggested that the PSS was elevated after adapting to the increased gain, compared to the reduced gain condition ($t_{17} = 3.12$, $p = 0.02$, $Cohen's\ d = 0.74$), consistent with our expectation that the PSS would shift in the direction of gain. However, no significant differences were found between gain of 1 and either non-unity gain blocks (reduced versus normal gain: $t_{17} = -0.78$, $p = 1.00$, $Cohen's\ d = -0.18$ and increased versus normal gain: $t_{17} = 1.75$, $p = 0.29$, $Cohen's\ d = 0.41$). Paired t-tests were then performed between the estimated parameters obtained from separate psychometric functions fit to data from the first two versus the last two test segments within the same adaptation blocks (test-retest). The results in Table 2 showed no significant test-retest differences for PSS or JND after any adaptation gain. This indicates that the absence of perceptual threshold shifts between non-unity and unity gains was not due to an initial adaptation followed by a decay of the effect over the test duration.

To determine if binocular information (e.g., convergence, disparity) contributed to the shift in PSS between the non-unity gains, we tested observers monocularly in Experiment 1b. There was no significant difference in PSS, JND, or head motion extent between the two viewing conditions (paired t-tests, $p > 0.05$). The comparison between Experiment 1a and 1b showed that binocularity neither attenuated nor enhanced the recalibration of perceptual thresholds. However, we did find a significant PSS effect and a significant difference between the most extreme gain adaptation conditions (0.667 vs. 2.0) under binocular but not monocular viewing, likely due to added depth information provided by stereopsis. ANOVA analyses of Experiment 1b showed no effect of gain adaptation under monocular viewing (Table 1), suggesting that the thresholds and head motion extent did not scale after adapting to monocular gain distortion.

To confirm that the overall lack of gain adaptation was not due to a failure to perform the required movements, we confirmed that observers moved as expected. To do so, task success rate was computed by taking the ratio of successful intersections with game targets divided by all task outcomes. ANOVA with task success rate as the response variable and gain as the independent variable showed no main effect, confirming that observers' game performance did not differ across gain conditions ($F_{2,34} = 1.12$, $p = 0.34$ for both viewing conditions). ANOVA on PSS with block order as the independent variable found no carryover effect under either viewing condition ($F_{2,34} = 0.89$, $p = 0.42$, and $F_{2,34} = 1.11$, $p = 0.34$, respectively).

We hypothesized that the lack of change in thresholds could be due to the unusual nature of the stimulation, as lateral motion is typically less common than forward motion. As a result, it may not drive adaptive processes as efficiently. In the next experiment, we evaluated this possibility using a more natural adapting stimulus and measured observers' PSS during back-and-forth sway while performing the same discrimination task.

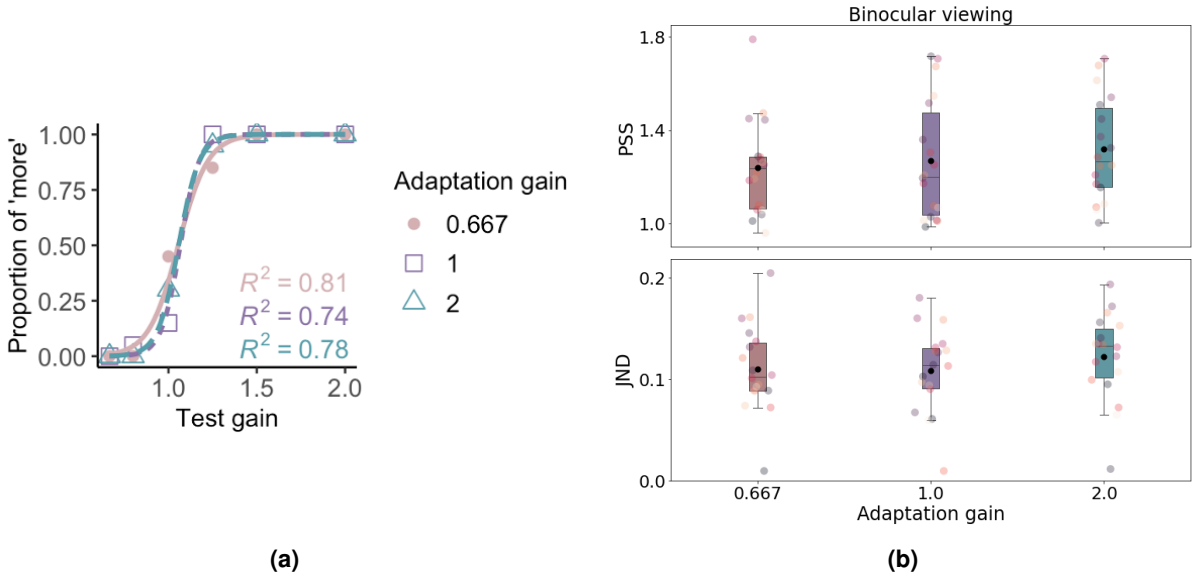


Figure 3. Perceptual thresholds in Experiment 1a under binocular viewing. (a) psychometric curves from a representative observer, responses at each test gain level are shown as dots, (b) estimated PSS (top) and JND from all observers. Box plots depict the middle two quartiles around the median (horizontal line) and mean (black dot), whiskers show 1.5 times the interquartile range (IQR) and individual data are shown as translucent dots.

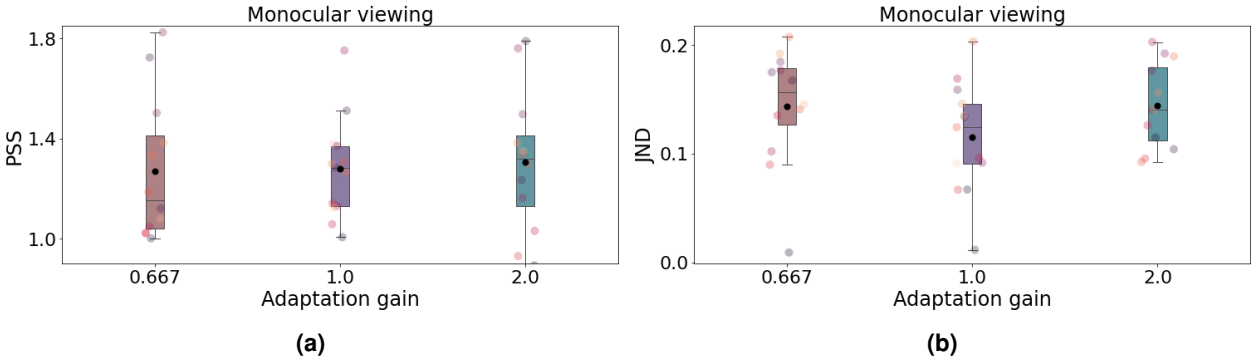


Figure 4. Estimated PSS (a) and JND (b) from all observers after adapting to the three gain levels in Experiment 1b under monocular viewing. The graphing convention follows that described for Figure 3b.

Experiment	*Measure*	*DFn*	*DFd*	*F*	*p*	η^2
1*a*	*PSS*	2	34	3.30	**0.05**	0.02
	JND	2	34	1.38	0.27	0.02
	Head motion extent	2	34	1.54	0.23	0.02
1*b*	*PSS*	2	34	0.67	0.52	4.00×10^{-3}
	JND	2	34	2.27	0.13	0.07
	Head motion extent	2	34	0.03	0.97	3.99×10^{-4}

Table 1. Results of ANOVA analyses on thresholds and head motion extent in Experiment 1.

Measure	*Adaptation gain*	*t*	*df*	*p*	*Cohen's d*
PSS	0.667	-1.18	17	0.25	-0.21
	1	-0.06	17	0.96	-0.01
	2	-0.97	17	0.35	-0.13
JND	0.667	0.17	17	0.87	0.04
	1	-1.59	17	0.13	-0.37
	2	-1.26	17	0.23	-0.44

Table 2. Results of paired t-tests on the test-retest (first two versus last two test segments) within each adaptation block of Experiment 1a. Tukey's Honestly Significant Difference adjustment was used to correct for multiple comparisons.

Experiment 2

The results of Experiment 1 suggested that observers did not recalibrate perceived motion under monocular viewing. To avoid the potential contribution of stereopsis and simplify the problem formulation, the adaptation and test phases were performed monocularly in Experiments 2 and 3.

The general design was the same as the previous experiment, but the gain in this experiment was applied to all translational directions, rather than only scaling and requiring lateral motion. As observers performed forward-direction motion most frequently during the adaptation phase, we changed the self-motion type to front-back in the test phase.

Methods

Participants

Eighteen observers (eleven female and seven male, age range 21–38 years, mean = 25.5) participated in this experiment. All participants had normal or corrected to normal vision. Among them, nine were experienced psychophysical observers (six of them participated in Experiment 1) and nine were novices.

Apparatus

This experiment requires natural and free motion, so virtual simulations were developed as a stand-alone application in C-Sharp using Unity 2022 and carried out on the Meta Quest Pro headset. The headset contains a pair of LCD displays with a resolution of 1800 × 1920 pixels per eye. The frequency of motion tracking and synchronous frame rate was 72 Hz. The horizontal and vertical field of view were 106°and 96°, respectively. The device comes with a removable light-blocking face mask insert, which was worn in these experiments to block the view of the external world around the edge of the display and provide immersion. The right-hand controller was used for the same functions as in Experiment 1. The left-hand controller was used as experimenter controls in case of disorientation during the session.

The virtual adaptation scene was constructed to match the geometry and approximate furnishings of the real lab in which the experiment was conducted, with desks, bookshelves, and a computer tower (Figure 5a). Task-related stimuli added included four stone pillars, each marked with a different two-dimensional (2D) shape (square, circle, pentagon and triangle), and four three-dimensional (3D) extruded versions of the same shapes: cube, cylinder, pentagonal and triangular prisms, and four wood stands that support the 3D objects (Figure 6).

Design and Psychophysical Procedure

The independent variables and their levels were the same as in Experiment 1. Rather than the method of constants used in Experiment 1, we used a pair of staircase procedures to estimate the PSS and JND. Transformed up-down staircase procedures[32] were used — a 2-up/1-down staircase interleaved with a 2-down/1-up, theoretically converging on the 70% and 30% points on the psychometric curve, respectively. For example, in the 2-up and 1-down staircase, the gain was increased after two

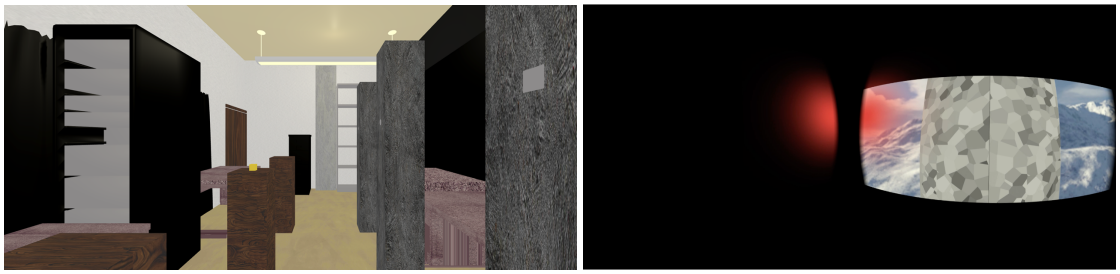


Figure 5. VEs used as adaptation and test scenes. (a) the virtual room with adaptation stimuli, (b) both eyes' views in the test phase, red boundary light denotes that the movement has met the forward distance requirement.

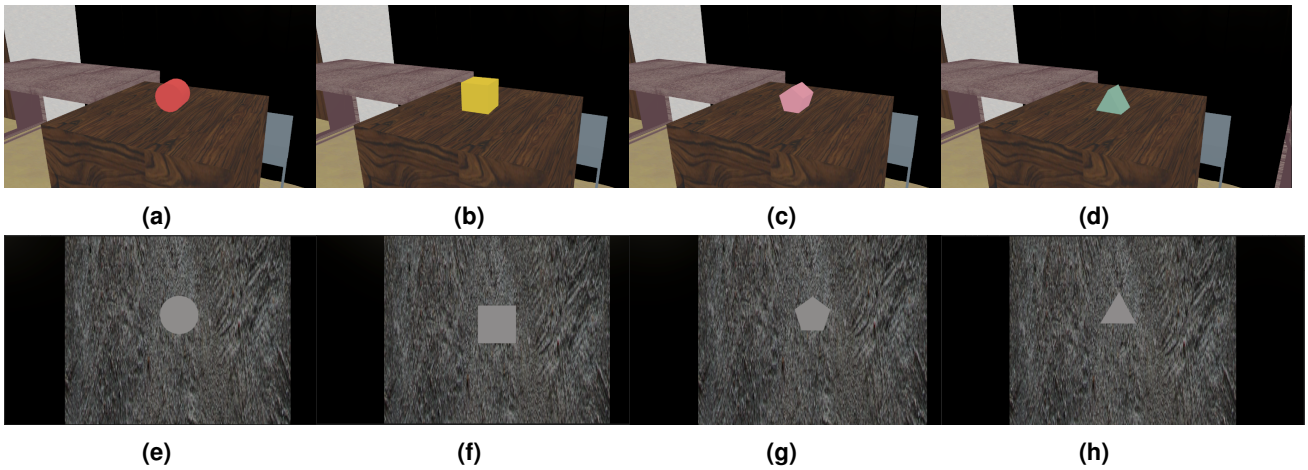


Figure 6. Visual stimuli in the adaptation phase. (a)–(d) 3D objects of different shapes on wood platforms, (e)–(f) 2D markers on destination stone pillars

consecutive responses of 'less gain' than expected and decreased after a single response of 'more gain'. The ending criterion of each staircase was set as either 10 reversals or 120 trials, whichever occurred first.

Several observers from Experiment 1 reported that they experienced aftereffects upon finishing the experiment including some disorientation and instability. To measure this, we added the fast motion sickness (FMS) survey as a dependent variable, assessed at the end of each block. The FMS is a single question motion sickness probe that asks observers to rate their current level of motion sickness on a 21-point scale from no sickness (0) to frank sickness (20)[33].

Experimental Procedure

The same procedure as in Experiment 1 was followed. Observers started with a 10-minute initial adaptation, followed by 2-minute testing segments interleaved with 2-minute top-up adaptation segments until the stopping criteria of both staircases were met. At the start of each adaptation segment, a single instance of one of the 3D objects was randomly selected and appeared on one of the wood stands. Simultaneously, each of the 2D shapes appeared on one of the pillars in random order. The task of the observer was to collect each object as it appeared and then find and move it to the matching receptacle shape on the appropriate pillar. Observers picked up the objects using the grip button on the right-hand controller and the object disappeared when it touched with the appropriate shape on the pillar. This marked a successful match, so a new random object and random set of receptacle locations appeared and the task was repeated (Supplementary Video S4). The purpose of these features was to provide an engaging and active task requiring controlled but natural motion during the adaptation phase. As a result, observers walked continuously to pick up the 3D objects and move them to the corresponding 2D shapes on the posts. They were instructed to complete the task as accurately and quickly as possible. In the test phase, the same fold stimulus, task and procedure from Experiment 1 were used, but observers moved their bodies forward and backward instead of laterally (Figure 5b).

Results

Figure 7 shows the staircase data from a reduced gain block for a representative observer. The PSS per adaptation block was estimated by averaging the 30% and 70% thresholds obtained from the convergence levels of the two staircases; the JND was estimated by scaling the 70% threshold to 75% then subtracting the PSS (Figure 8). One-way repeated-measures ANOVA with PSS and JND as the response variables and adaptation level as the independent factor showed no significant effect of adaptation gain (Table 3). Pairwise t-tests did not show the hypothesized significant increase or decrease in PSS or JND relative to unity gain after adaptation to high and low gain, respectively (Table 4). ANOVA of the FMS ratings (Figure 9) showed that adaptation gain had a large effect (Table 3), and both non-unity gain conditions showed a significant increase compared to the gain of 1 condition ($p \leq 0.01$, $Cohen's\ d > 0.8$, see Table 4). As in the previous experiment, there was no effect of block order on PSS, JND, or task success rate. These results suggest that while our gain manipulation did not induce perceptual adaptation, it may have impacted comfort and postural control.

Measure	*DFn*	*DFd*	*F*	*p*	η^2
PSS	2	34	0.08	0.96	4.95×10^{-4}
JND	2	34	0.34	0.72	0.02
FMS	2	34	10.78	$\mathbf{3.66 \times 10^{-4}}$	0.23

Table 3. Results of ANOVA analyses on the PSS, JND, and FMS in Experiment 2.

Measure	*Comparison*	*t*	*df*	*p*	*Cohen's d*
PSS	0.667 versus 1	-0.25	17	1.0	-0.07
	0.667 versus 2	-0.16	17	1.0	-0.04
	2 versus 1	-0.14	17	1.0	-0.04
JND	0.667 versus 1	-0.71	17	1.0	-0.19
	0.667 versus 2	-0.69	17	1.0	-0.19
	2 versus 1	0.02	17	1.0	6.24×10^{-3}
FMS	0.667 versus 1	4.06	17	**0.01**	0.94
	0.667 versus 2	-1.38	17	0.57	-0.32
	2 versus 1	5.30	17	$\mathbf{1.00 \times 10^{-3}}$	1.25

Table 4. Results of pairwise t-tests on the PSS, JND, and FMS in Experiment 2. Tukey's Honestly Significant Difference adjustment was used to correct for multiple comparisons.

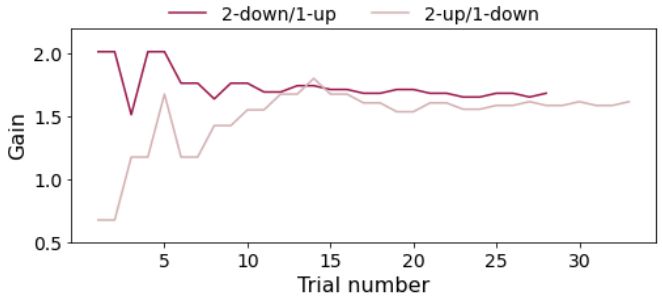


Figure 7. Sample staircases from a reduced gain adaptation block for one representative observer in Experiment 2 are shown here.

Experiment 3

The results of our previous experiments indicated that adaptation to non-unity visual gain did not significantly alter perceived motion, yet resulted in higher FMS ratings in Experiment 2. This discrepancy raised the possibility that adaptation effects might manifest more prominently in balance control systems rather than in conscious motion perception. To evaluate the hypothesized effect of adaptation on postural control, in this study, we used the same adaptation paradigm but measured the

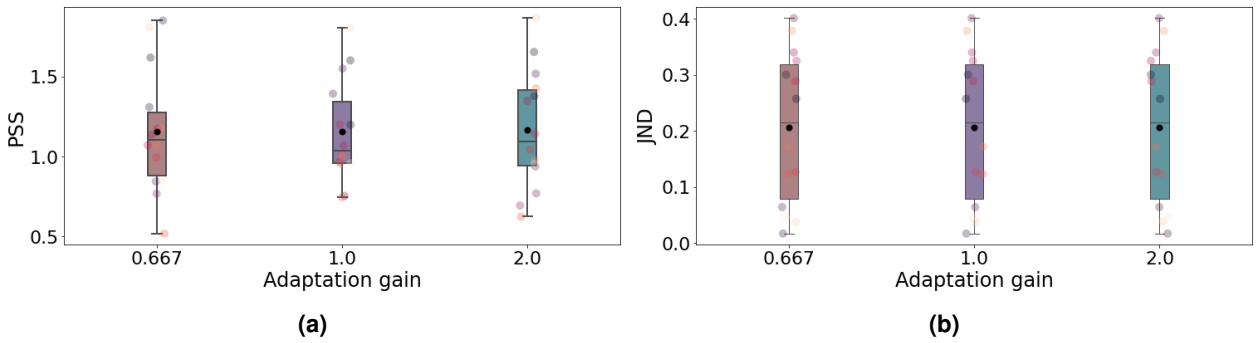


Figure 8. Estimated PSS (a) and JND (b) from all observers after adapting to the three gain levels in Experiment 2. The graphing convention follows that described in Figure 4.

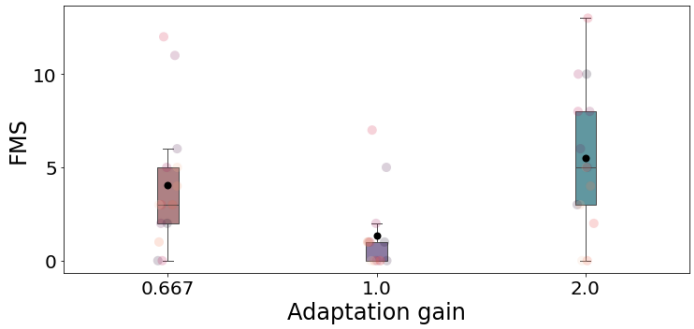


Figure 9. Results for FMS ratings after adapting to the three gain levels in Experiment 2. The graphing convention follows that described in Figure 4.

effect of gain adaptation on postural control mechanisms in Experiment 3. We evaluated left-right and front-back motion adaptation in Experiment 3a and 3b, respectively.

Methods

Participants

Eighteen observers (nine female and nine male, age range 19–39 years, mean = 26.3) participated in this experiment. All participants had normal or corrected to normal vision. Among them, ten also participated in either Experiment 1 or 2, thirteen were experienced psychophysical observers, and five were novices.

Design

The independent variables were adaptation gain (0.667, 1.0, and 2.0), visual condition during test phase (visual stimulus present and absent), motion direction (front-back and left-right), with 4 repeats of each condition. Dependent variables were head motion (balance control) and FMS scores.

Procedure

The apparatus, the adaptation VE stimulus, and the adaptation procedure were the same as in Experiment 2. There were two major changes to the test procedure: (a) the 2-minute test segments were divided into two 1-minute halves and (b) we added a baseline test-phase condition for comparison, which assessed stability performance without any prior adaptation. Two virtual environments were used in the test phase. In one, the scene was blank and only a white ground bar was visible (visual stimuli absent). In the other, the virtual room was visible without the adaptation-related platforms, shapes and pillars, with the ground bar indicating the test position. Upon finishing each adaptation segment, observers were transitioned to one of the two test scenes in counterbalanced sequential order and they were instructed to walk to the ground bar. Once they were ready to maintain a quiet stance, they pressed a button and the ground bar disappeared. The task in the test phase was to maintain balance in Romberg stance for 1 minute, in which the inner sides of the feet stayed in contact (or 'feet together'). The visual stimulus present condition was similar to the 'moving room' stimulus in Lee and Lishman's studies[24,25]. The room oscillated sinusoidally at a frequency of 0.2 Hz along either the front-back (Experiment 3a) or the left-right (Experiment 3b) axes (Supplementary Video S5 and S6).

Results

On each trial, the first and last second of the recorded head motion data were discarded to eliminate transient onset and offset effects, leaving the stable-state response to the visual stimulus. The mean of the resulting head position in the direction of the current experiment (z for Experiment 3a and x for Experiment 3b) over the trial was subtracted to remove any systematic bias in starting position. To characterize the overall amplitude of postural variability, the resulting head position signal was then summarized by its root mean square (RMS) variation for both the visual stimulus present and absent conditions (Figure 10)[34]. For the visual stimulus present conditions, the head motion (position with time stamp) was further processed through a band-pass filter of 0.1–0.3 Hz to isolate frequency components of interest (0.2 Hz) and transformed into a periodogram to estimate its power spectral density (PSD). The PSD represents the distribution of variance, or signal power (note this is distinct from statistical power) across different frequencies and allows us to assess the coupling with the visual perturbation. Specifically, the power at 0.2 Hz was extracted from the PSD to estimate the sensitivity at the frequency of the moving room stimulus (Figure 11a). Linear mixed models (LMM) were used to analyze RMS and power at 0.2 Hz in Experiment 3a and 3b, with observer treated as a random effect. The general model specification was:

$$D.V. \sim Adaptation + (1|Observer) \tag{2}$$

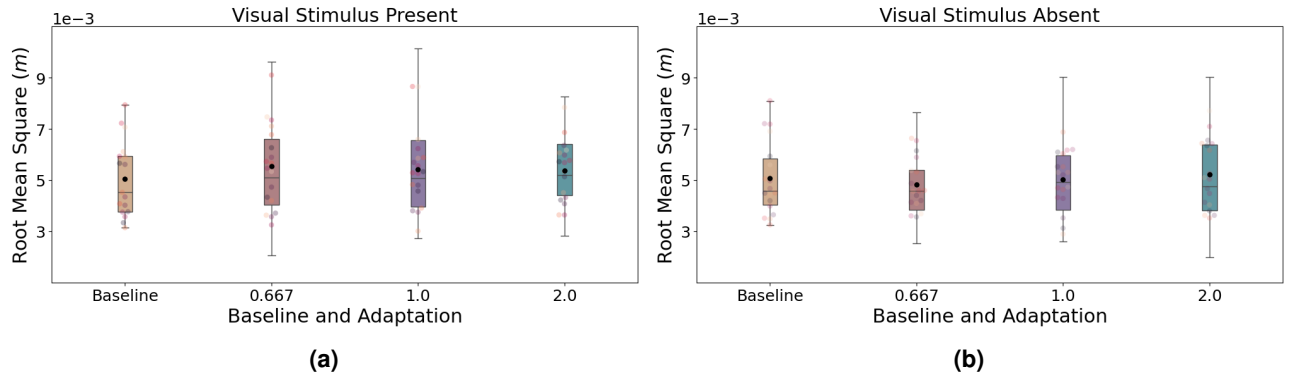


Figure 10. Root mean square (RMS) postural variability of baseline and adaptation blocks Experiment 3a. (a) visual stimulus present, (b) visual stimulus absent. The graphing convention follows that described in Figure 4.

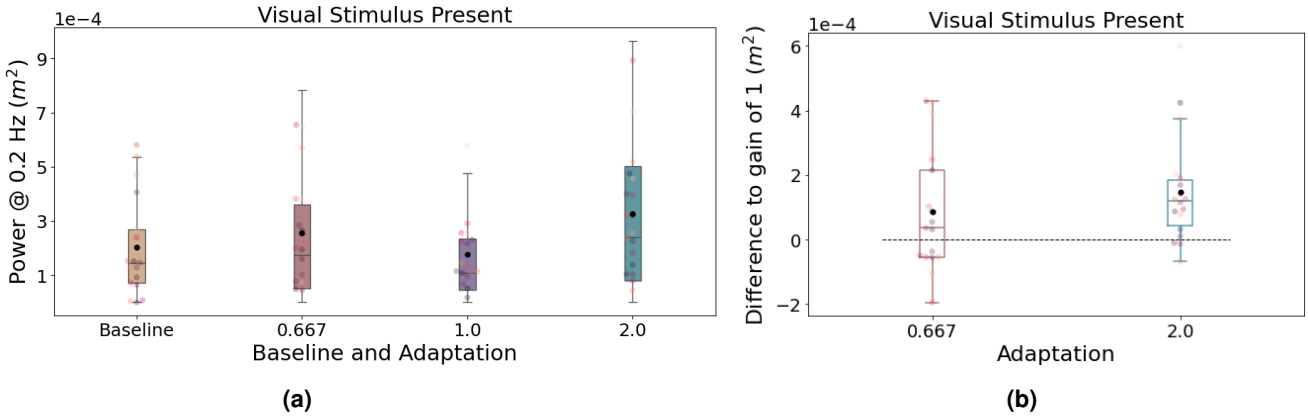


Figure 11. Power spectral analysis for Experiment 3a. (a) power at 0.2 Hz (m^2) of both baseline and adaptation blocks in visual stimulus present conditions, the graphing convention follows that described in Figure 4, (b) difference scores of power for non-unity gains, all values are relative to each observer's power at a gain of 1 (the dashed line).

For robust estimates of the effects, we identified and removed individual outlier data points 1.5 times the interquartile range (1.5·IQR) above the 75% quartile or 1.5·IQR below the 25% quartile shown in Figures 10 and 11. When removing outlier data we retained the other data points for a given observer (i.e., we did not exclude the observer). After removing outliers, the front-back moving room results of Experiment 3a showed a significant main effect of adaptation gain on both RMS and power (Table 5). The effect size of the adaptation gain was small to medium for RMS and medium-to-large for power

Experiment	*Measure variable*	*NumDF*	*DenDF*	*F*	*p*	η_p^2	R_m^2 (%)
3*a*	RMS	2	179.28	3.05	**0.05**	0.03	1.4
	Power	2	179.63	11.50	$\mathbf{2.00 \times 10^{-5}}$	0.11	7.3
3*b*	RMS	2	170.72	0.49	0.61	5.75×10^{-3}	0.2
	Power	2	172.12	3.51	**0.03**	0.04	2.5

Table 5. Results of LMM analyses on RMS and power in Experiment 3 with the visual stimulus present conditions. Denominator degree of freedom was Satterthwaite's approximation.

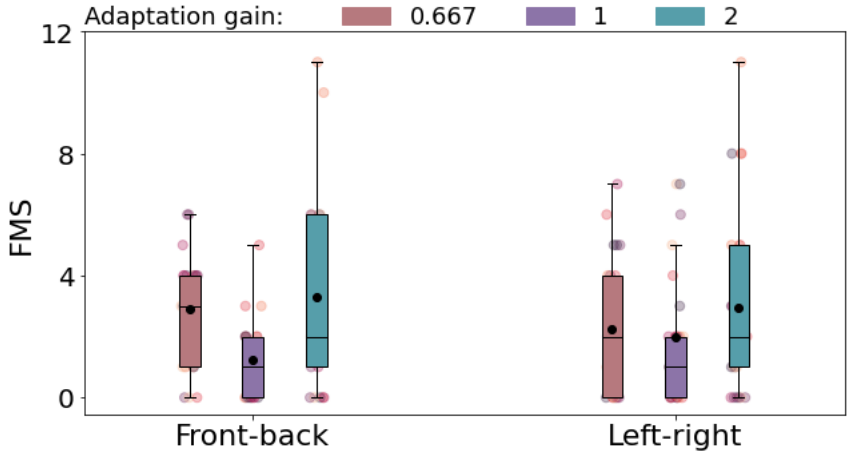


Figure 12. FMS ratings of front-back and left-right experiments. The graphing convention follows that described in Figure 4.

($\eta_p^2 = 0.03$ and 0.11, respectively), given the large random effect reflecting individual differences in power ($R_m^2 = 1.4\%$ and 7.3%). We expected that the increased gain would serve as to amplify visual feedback to postural stability regulating balance control, whereas reduced gain would weaken postural self-regulation as reflected in both RMS and power. However, pairwise comparisons showed that increased gain produced a small to medium elevation of RMS and a large elevation of power (*Cohen's* $d = 0.42$ and 0.81), compared to normal (2 versus 1); adaptation to reduced gain did not show the expected increase compared to normal (0.667 versus 1), the only difference was between this and the power in the increased gain condition (Table 6). The analysis on difference scores of power at 0.2 Hz (Figure 11b) indicates that observers exhibited relatively consistent responses following the high gain adaptation (markedly more clustered around the mean), suggesting a selective increase in power at high gain during the front-back motion elicitation. The large random effect of 7.3% was primarily driven by the relatively broad spread in difference scores observed at the low gain level in Figure 11b.

These results run counter to our expectation that postural sway amplitude would show a decreasing trend as a function of increasing gain. In Experiment 3b (left-right axis), unlike Experiment 3a, there was no significant effect of adaptation on RMS, and the elevation in the power at 0.2 Hz for non-unity versus unity gain was less pronounced (Table 5). Post-hoc analysis showed that this effect was apparent between adaptation gains of 0.667 and 1 (Table 6). For the visual stimulus absent conditions, while there was no stimulus present to drive sway, participants could have exhibited changes in stability. If the results in visual stimulus present conditions were replicated, the adaptation would be seen in elevated RMS in altered gains compared to normal gain. LMM on RMS measured in the dark found no significant effect on the adaptation blocks in Experiments 3a or 3b. However, two-sided pairwise t-tests between baseline and individual means of normal gain blocks suggested normal gain produced an increase in RMS only along the left-right axis ($t_{17} = 2.36$, $p = 0.03$, *Cohen's* $d = 0.56$). We argue that this effect was not due to adaptation as the spontaneous postural sway should not depend on test direction, given the dark scene contained no directional component.

These results show that adaptation to increased gain is associated with increased visually-elicited postural responses along the front-back direction, while reduced gain produced a small decrease in synchronous postural sway in the left-right direction. For motion sickness symptomatology, FMS ratings were significantly higher in both non-unity gain blocks in Experiment 3a, compared to gain of 1, whereas in Experiment 3b, gain did not significantly affect FMS (Figure 12 and Table 7). LMM on postural response and FMS found no significant effect of testing order in either experiment, suggesting the observed effect of gain adaptation was not due to the buildup of discomfort or fatigue over blocks. The analysis of task success rate showed no effect of gain level or block order, confirming that the observed effects were due to gain adaptation rather than task difficulty or

Experiment	*Measure variable*	*Comparison*	*t*	*df*	*p*	*Cohen's d*
3a	RMS	0.667 versus 1	0.97	181	0.59	0.17
		0.667 versus 2	-1.42	181	0.33	-0.25
		2 versus 1	2.45	180	**0.04**	0.42
	Power	0.667 versus 1	2.01	182	0.11	0.36
		0.667 versus 2	-2.65	181	**0.01**	-0.46
		2 versus 1	4.77	181	$< \mathbf{1.00 \times 10^{-4}}$	0.82
	FMS	0.667 versus 1	2.75	17	**0.04**	0.60
		0.667 versus 2	-0.38	17	1	-0.09
		2 versus 1	2.67	17	**0.03**	0.63
3b	RMS	0.667 versus 1	-0.53	172	0.86	-0.10
		0.667 versus 2	-0.99	171	0.58	-0.18
		2 versus 1	0.47	172	0.89	0.08
	Power	0.667 versus 1	-2.58	174	**0.03**	-0.46
		0.667 versus 2	-1.83	171	0.16	-0.33
		2 versus 1	-0.75	173	0.74	-0.13
	FMS	0.667 versus 1	0.32	17	1.0	0.08
		0.667 versus 2	-0.85	17	0.98	-0.20
		2 versus 1	1.15	17	0.57	0.27

Table 6. Partial results of pairwise comparisons on RMS (visual stimulus present), power (visual stimulus present), and FMS in Experiment 3. The Kenward-Roger method was used to approximate degrees of freedom, Tukey's Honestly Significant Difference adjustment was used to correct for multiple comparisons.

Experiment	*DFn*	*DFd*	*F*	*p*	η^2
3*a*	2	34	6.97	$\mathbf{6.00 \times 10^{-3}}$	0.14
3*b*	2	34	2.82	0.07	0.03

Table 7. Results of ANOVA analyses on FMS in Experiment 3.

failure to adapt movements under varying gain.

General Discussion

Dissociation of perceptual and motor responses

Our results demonstrate a clear distinction between the perceptual and motion gain adaptation: while postural sway exhibited significant gain-dependent adaptation, the perceptual estimates of PSS under active self-motion did not demonstrate the predicted adaptation after exposure to higher or lower gain. We predicted that if observers fully adapted, estimated PSS would match the adaptation gains of 0.667 and 2 in reduced and increased blocks, respectively. However, comparison with Figures 4a and 8a clearly shows that there was little adaptation for either the lateral or more natural adaptation movements in Experiment 1 or 2. The PSS estimates were relatively flat across gain levels and changes were much smaller than predicted for full adaptation. The confidence intervals on the PSS were on the order of 0.1, which should provide enough sensitivity to detect shifts of the expected magnitude. The measures were also stable, as reflected by the similar PSS and JND outcomes across experiments. Further, the data of repeated observers were compared between our previous and current studies, and a high test-retest correlation was found between the two points in time (Pearson correlation coefficient, $r \geq 0.9$ for both lateral and front-back thresholds)[13]. These results confirm that the weak or null adaptation effects in Experiments 1 and 2 were not due to ceiling/floor effects of the measurement approach. Interestingly, the PSS estimates were lower and more accurate (closer to 1) in Experiment 2 than Experiment 1, suggesting that the more natural adaptation and test motion in the second experiment did improve performance. Despite not showing adaptation of the PSS, observers reliably discriminated the levels of test gain, with reasonable JNDs. This confirms that the observers understood the task and could perform the judgments reliably across gain and motion conditions.

Our adaptation protocol was able to produce postural effects but not perceptual adaptation. The initial adaptation period of 10 minutes was based on expected time course of sensory adaptation, while managing the experiment length and avoiding over-fatiguing the participants. Given that the initial adaptation period was followed by three 2-minute "top-up" adaptation periods, the adaptation duration was 16 minutes in total. While longer adaptation periods may have produced stronger effects,

the motion aftereffect reportedly shows a square-root dependence on adaptation time and typically begins to saturate after exposures on the order of 1 minute[35,36]. In short-term vestibulo-ocular reflex (VOR) adaptation paradigms, the reflex is typically well adapted at the end of the 15-min sequence and shows continuous adaptation throughout[37]. Some complex forms of sensorimotor rearrangement can take much longer to manifest[38], but these cases often involve motor learning, developing explicit strategies, or mechanisms other than perception.

We expected that after 10 minutes of adaptation to a given gain in the current study, the PSS would show a considerable shift toward the full adaptation prediction and that the JND would scale proportionally to the shift in PSS. The lack of perceptual effects suggests that this adaptation did not occur or that it was fragile and short-lived. One possibility is that there was adaptation but it was 'washed out' by the relatively large-scale active motion in the test phase, the additional vestibular and proprioceptive cues from which may have quickly recalibrated the perceptual system. This dynamic updating is consistent with previous evidence showing that active self-motion engages predictive mechanisms and efference copy signals that stabilize perception and resist sensorimotor remapping in altered environments[22]. Recent studies also suggest that head and body movements are likely to contribute to self-motion perception[2], which applies to the test phase in our Experiment 1 and 2 but less so to Experiment 3. While we did not find evidence of an effect of adaptation on the perceptual judgments, balance was affected. This contrasts with the literature on sensorimotor adaptation in reaching and locomotion, where perceptual and motor recalibrations co-occur[17]. According to our 'active movement' hypothesis, the adaptation seen in the postural measures persists because the small compensatory movements during quiet stance were not sufficient to produce the rapid recalibration.

An alternative explanation for the perceptual/motor dissociation is that the sensory processing itself was not affected by adaptation but only the motor responses were adjusted. This explanation is consistent with prior models positing separate pathways for action (dorsal stream) and perception (ventral stream) in the neocortex[39]. Our proposed explanation is also aligned with subcortical mechanisms that continually tune postural and oculomotor reflexes to maintain stability in response to perturbations. Examples include vestibulo-ocular and vestibulo-spinal reflexes that drive short latency motor responses to afferent vestibular signals, but that also maintain their accuracy by modulating their response to visual and proprioceptive error signals. Lavalle and Cleworth found that a higher gain such as 16 might be maladaptive to upright stance and elicited postural instability[40]. These adaptive mechanisms involve the cerebellum, vestibular nuclei and other structures[41,42] which drive adaptation in response to descending visual input but do not directly influence visual perception. It is also possible that the presence of adaptation in postural stability without accompanying perceptual calibration suggests that balance control may act as an early compensatory filter for resolving sensory mismatches before perceptual awareness is engaged[43].

Postural Adaptation

In Experiment 3, we predicted that postural sway during quiet stance would decrease with increasing adapting gain, as in previous studies[34,44,45]. However, our results showed that, under visual perturbation, overall RMS variability and synchronous postural sway (power at 0.2 Hz) increased following front-back adaptation to the high gain, and decreased for lateral perturbation following low gain.

It is unclear why this pattern of results occured. We speculate that this might reflect recalibration of the expected visual consequences of normal postural movements: the sensory system could have recalibrated expectations to match the modified optical flow manipulated by gain as a 'new normal'. Then, during testing, the motor system would need to produce the head motion that matches the expected amplitude of the visual signal. Consistent with this proposal, the power at 0.2 Hz in Experiment 3a showed significant elevation following adaptation to the high gain level, compared to both unity and low gains. In Experiment 3b, the effect size was notably less, and rather than an increase at high gain, the power modulation in 3b emerged as a reduction at low gain compared to unity gain. That is, synchronous postural sway was modulated by the direction of gain manipulation. Following the same pattern, adaptation to high gain with front-back motion perturbation increased RMS postural variability to match the recalibrated perception of a stable world, althought this was the only significant difference across all the pairwise comparisons of RMS in Experiments 3a and 3b. The direction of the significant change in power and RMS agrees with the direction of gain reduction/amplification in Experiments 3a and 3b, although whether it was the increase or decrease that was significant differed in the two experiments.

This interaction between the directions of visual perturbation and gain manipulation found in power and RMS shows that effects of gain adaptation are not evenly distributed. Sawaguchi's study in 2022 suggests that vision plays a critical role in returning the body to the neutral position, particularly along the anterior–posterior axis[46]. They found that when visual input was removed, the peak displacement of the center of pressure (COP_{peak}) increased in the anterior–posterior direction, indicating a delayed recovery from sway. This supports the hypothesis that visual feedback contributes specifically to ankle-mediated postural corrections in the sagittal plane (front-back direction), which are essential for maintaining balance during quiet stance[47,48]. It is possible that the relative insensitivity to increased gain in the left-right direction in our experiment is due to the configuration of joints in the lower extremities[47,49]. Another explanation for the lack of RMS modulation in Experiment 3b is that observers adapted more effectively to forward motion due to task demands and the ecological relevance to locomotion and

walking. Consequently, balance control along the lateral direction was less biomechanically affected as long as visual input was available. In contrast, in visual stimulus absent conditions, the insensitivity to lateral motion gain was not interpreted as visual feedback being unavailable to inform stability, but rather as postural variability being independent of gain[50]. In fact, the LCD screens of the headset could not present complete darkness. However, the dimly visible fixed edges of the display field of view were constant and did not change in response to head motion; thus, this 'open-loop' stimulus could not be used for effective balance control.

Implications of visuomotor adaptation for motion sickness

The FMS ratings showed elevations following non-unity gains in both Experiments 2 and 3, especially when tested along the front-back direction in Experiment 3a, matching the predominant direction of experienced motion gain. This is consistent with Kim et al.'s finding that the perceived scene instability always increases with the degree of linear visual-vestibular conflict[51]. Visually-induced motion sickness (VIMS) has been linked to conflicts between expected and actual non-visual sensory input and visually-mediated self-motion perception, such as arise during mismatched gain[2,52]. Prior work has found that repeated exposure to gain-manipulated VR environments can decrease VIMS susceptibility over time[53–56]. This may explain why FMS did not change significantly between gain conditions when elicited by left-right motions: observers had limited exposure to lateral motion manipulation in Experiment 3b, potentially leading to insufficient sensory conflict and minimal prior expectations that could elicit significant discomfort. This supports the assumption that the sensorimotor system is likely recalibrating constantly to reduce conflict, which could be a potential mechanism to mitigate discomfort[57].

The distinction between perceptual and motor responses as well as the elevation in FMS reported here have implications for understanding adaptation scenarios in other experiments. A study by Kasegawa et al.[52] reported that appropriately scheduled time intervals between VR sessions significantly reduced VIMS and alleviated motion sickness symptoms. Adaptation to VOR could be retained for longer than the training duration or outside of the training context[37,58], and recalled upon re-entering VR[55]. These studies suggest that the brain treats a VE as a distinct 'context', if so participants likely 'save' a motor memory that leads to direction-specific sensorimotor adaptation which does not washout outside the adaptation-phase context. In Experiment 3a, the power of visually-elicited postural sway was amplified after adapting to increased gain (under front-back visual perturbation), while in Experiment 3b the power decreased after adapting to reduced gain (under left-right visual perturbation). These may result from the corresponding motion gain in the previous adaptation phase, potentially triggering a 'recall' of the context saved during training. As a result, the motor system may have engaged postural sway accordingly, matching the expected amount of visual feedback to maintain perceptual stability.

Recent findings suggest that individual differences in visual sensitivity, particularly to motion parallax cues, can predict susceptibility to motion sickness in VR environments[59,60]. Fulvio et al. demonstrated that observers who are more sensitive to motion parallax cues are more likely to experience greater discomfort when those cues are absent or unreliable[59]. This supports the cue conflict theory that individuals who are more attuned to specific sensory cues are more affected by their absence or inconsistency, leading to increased discomfort. Bunday and Bronstein[61] demonstrated that adaptation to discordant visual-vestibular motion cues influences locomotor behavior, suggesting that sensory strategies are continuously weighted and updated based on environmental conditions. These findings underscore the need for adaptive VR systems that account for individual variability in sensory processing. Importantly, VR serves as a flexible tool for systematically manipulating visual conflicts, but the neural mechanisms underlying sensory adaptation and plasticity are general-purpose and not necessarily VR-specific. For example, visuomotor gain changes typically occur when exposed to magnifying or minimizing lenses, and in the context of VOR have been shown to be adaptable in both virtual and real environments[62,63]. Future work could elaborate on individual differences in self-reported motion sickness, contributing to the development of responsible and consistent VR systems that enhance user comfort and safety.

Conclusions

Our experiments show that relatively brief exposure to virtual motion with altered visual gain leads to systematic changes in quiet stance postural responses. While maintaining stability during front-back visual stimulation was challenging, motor responses were less sensitive to lateral visual perturbation. Motion sickness ratings were elevated after adapting to altered gain when tested with a front-back moving room stimulus. This may be because observers most commonly made and experienced forward compared to lateral movements during natural motion adaptation, leading to heightened sensitivity to front-back visual perturbations. Our findings suggest that gain adaptation in the VE reflects the recalibration of the motor responses of self-motion but not perception, suggesting that perceptual recalibration operates independently of motor adaptation. This dissociation informs theoretical models of multisensory integration and has practical implications for future interface designs. Fall risk assessment and rehabilitation systems should consider this asymmetry to increase the fidelity, implementing directional, perturbation-based balance training programs. Given the correlation between gain and the directional selectivity of motor adaptation, individuals with compromised sensory function (e.g., hearing loss, mobility challenged) might benefit from balance

training under reduced-gain lateral visual elicitation and increased-gain front-back visual elicitation, enhancing reactive postural control.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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Author contributions statement

L.W. and R.A.: conceptualization, methodology, writing – review and editing, supervision, project administration, funding acquisition.

X.T.: conceptualization, experiment implementation, methodology, formal analysis, investigation, visualization, writing – original draft, writing – review and editing.

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